PROJECT REQUIREMENTS PDF

Hospital Readmission Risk Prediction Project

A Machine Learning – Based Decision Support System

Item	Detail
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Faculty guide	*[Guide name]*
Status	In progress

1. Introduction

1.1 Purpose

This document specifies the requirements for a machine learning based decision support system. Which estimates the likelihood of the patient who is already discharged and gets confused whether to get readmitted again or not after a few days. The main purpose of this project is to guide the discharged patients based on their condition and situation, whether to get readmitted or not.

1.2 Problem Statement

Hospitals often face challenges in identifying patients who are at a high risk of readmission after discharge. Develop a Machine Learning-based decision-support system that uses historical, de-identified patient records, including demographic details, previous hospital visits, diagnoses, treatment information, and length of stay, to predict the likelihood of readmission within a defined period. The system should identify the factors associated with readmission, compare model performance across relevant patient subgroups, and highlight high-risk patient profiles that may benefit from additional follow-up care.

1.3 Scope

In scope: Data preparation, model training and evaluation, identification of risk factors, high-risk profile discoveries, a REST API (FastAPI), a web user interface, testing and local or cloud deployment with an access URL.

Out of scope: Integration with live hospital information systems, use of identifiable patient data, clinical diagnosis or treatment recommendations, and regulatory certification.

2. Requirements

2.1 Dataset selection

Selection of the right dataset to train the model is very much important in the model's performance. The dataset which gives the relevant information and contains the features which determines the prediction is to be selected appropriately. The model which we are going to develop is going to be trained using the dataset **diabetes_data.csv**

Why choosing only diabetes dataset?

Firstly, it matches the project description almost exactly. The project asks for the demographics, previous hospital visits, diagnoses, treatment information, length of stay and readmission outcome. These features were spotted in the dataset. Secondly, it has ready-made well defined readmission target column ("readmitted" column).

It is de-identified and openly licensed; it is large enough for well training of the model.

Dataset: diabetes_data.csv

2.2 Required Tech Stack

Layer	Tech Stack
Frontend	HTML, CSS & JavaScript
API	REST API, FastAPI, JSON
Backend	Python 3, Uvicorn, Pydantic
Model Training	Scikit-learn
Data Preprocessing	Pandas, Matplotlib, Seaborn

Runs on a standard laptop (local) and on a free cloud tier.

2.3 Functional Requirements

The system shall accept patient inputs using a form in the web interface.

The system shall return the predicted probability of readmission within 30 days for a single patient.

The system shall assign each prediction to a risk tier (Low, Medium, High).

The system shall present model performance separately for age group, gender, race and primary diagnosis group.

The system shall display a disclaimer on every results screen stating that output is decision support only and not a diagnosis.

2.4 Git Repository

Git repository will be created and updated here later.

Notice for Team Members: This requirements PDF is of version v0.1 and will be updated eventually before the final project submit.